COMM 5559 HW 1 – Alteryx

Note to MSA students: You can reuse the workflow you created for GCOM 7380 and start with step j.

Accounting serves an important role in CEO contracting. In this exercise, we will examine the relation between firm accounting-based performance and CEO ***forced*** turnover. A CEO forced turnover occurs when a firm "fires" its CEO, instead of the CEO voluntarily resigning/leaving. People often expect firms to replace CEOs if they experience poor performance. Thus, there would be a negative relation between firm performance and CEO turnover. We will test this prediction using regressions.

To complete this exercise, you will need data from 2 sources. Both data files are provided on Canvas.

(1) firm financial data:
The firm financials dataset contains financial statement data of US public firms between 2010 and 2018. In particular, we have the following columns.
Gvkey: an ID that identifies firms;
Year: fiscal year;
Assets: total assets;
Liabilities: total liabilities;
Earnings: earnings before extraordinary items;
RD: research and development expenditures;
SIC: SIC code, which is used to identify industries.

Because this file contains data from multiple years, we will see repeated gvkeys, but they are not necessarily duplicates.

(2) CEO turnover data:
The CEO turnover dataset is collected and provided by Professors Florian Peters (University of Amsterdam) and Alexander Wagner (University of Zurich). The file contains CEO forced turnover data between 2010 and 2018. There are 3 columns in the file: gvkey, year, and turnover. Gvkey and year are defined as before. Turnover is an indicator of a forced CEO turnover (i.e., if a forced turnover happens, turnover = 1).

Thanks to Professors Peters and Wagner, this data file is "clean" and can be used directly. However, note that this data file only contains the firm-years when a forced turnover occurred (i.e., when turnover = 1). In other words, if there was no forced turnover during a specific firm-year combination (i.e., when turnover = 0), then that firm-year is **NOT** included in this data file (see the steps below for more detail).

Analysis:
Please complete the following steps in Alteryx.

a. Import both data files to Alteryx.

b. Use the Select tool to change data types as necessary. In particular, we need to make sure that the data type for numerical fields is "double." I also recommend setting the type of "turnover" to double. This is helpful for step h (the reason is discussed below in step h).

c. Let's focus on cleaning the firm's financial data first. In this step, let's remove the duplicates using the Unique tool. As mentioned above, records with repeated gvkeys are not necessarily duplicates because we can have data from the same firm for multiple years. Thus, using "gvkey" itself won't be sufficient to determine duplicates. Instead, we both "gvkey" and "year."

d. Next, let's filter the data by creating the following conditions: (1) Assets must be greater than 100 (note the numbers are in millions, so 100 means 100 million), (2) Earnings shouldn't be null, (3) Liabilities shouldn't be null, (4) SIC code shouldn't be null, (5) the first digit of the SIC code is **NOT** 6 (*Hint: the Startswith() function can check if the first digit is 6, e.g., Startswith([SIC],"6"), and the "not" operator in Alteryx is !*).

   Condition (5) removes all firms in the financial service industry. We remove them because financial service firms are heavily regulated, and the regulations can directly influence firm performance without the CEOs' control. In other words, their performance sometimes does not reflect CEOs' true ability/effort. Thus, our results could be biased if we do not exclude these firms.

e. Use the Data Cleansing tool to replace missing RD with 0. We will also remove all null rows using the Data Cleansing tool.

f. In this step, let's create the 4 variables used in the analysis, namely, return-on-assets, liability-to-assets ratio, and R&D-to-assets ratio. They are defined as earnings scaled by assets, liabilities scaled by assets, and RD scaled by assets.

g. Next, join the two data files. The type of join that we will use is called "left outer join" (assuming the financials data is the left input, and the CEO turnover data is the right input). That is, we will keep all records from the left input and then append them with (a subset of) the data from the right input. Below is a simplified example of a left outer join.

gvkey	year	ROA
1	2010	0.05
1	2011	0.10
2	2010	0.03
2	2011	-0.15

gvkey	year	turnover
2	2011	1
1234	2015	1



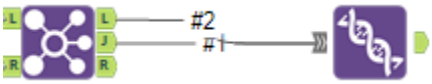
gvkey	year	ROA	turnover
1	2010	0.05	
1	2011	0.10	
2	2010	0.03	
2	2011	-0.15	1

   As seen, the output should contain all the records from the left input. However, if a firm-year only appears in the right input but not in the left input (e.g., the record with gvkey = 1234), then it will not be included in the output.

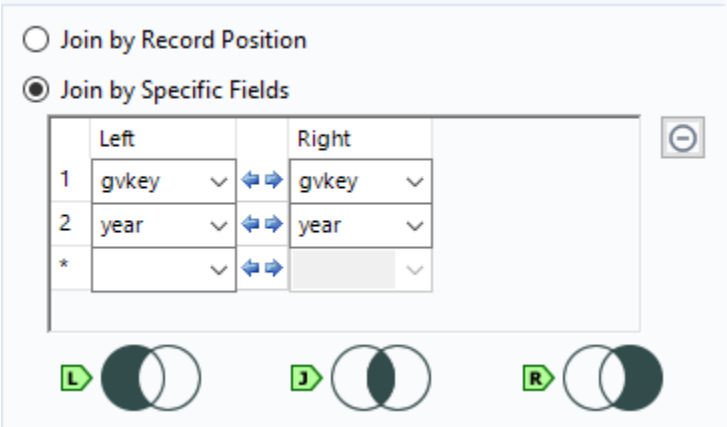
   While left outer join is not directly available in Alteryx, we can achieve this goal by combining a Join tool with a Union tool. Conceptually, left outer join is the combination of left unjoined and inner join. For example, in the example above, the first 3 rows of the output are essentially the left unjoined (i.e., records that only appear in the left input), and the last row is the inner join (i.e., records that appear in both inputs).

gvkey	year	ROA	turnover	
1	2010	0.05		Left unjoined
1	2011	0.10		
2	2010	0.03		
2	2011	-0.15	1	Inner join

Thus, to create a left outer join, we can **Union the L and J outputs of the Join** tool as below.



Also, to configure the Join tool, we need to use 2 fields, gvkey and year, because these two fields jointly identify the records. See below.



h. As shown in the example above, after joining the data files, we will have missing values for the turnover field. When the value is missing, it means that there was no forced CEO turnover. Therefore, we can replace the missing values with 0. (*Hint: if you use a Data Cleansing tool to replace missing values with 0, "turnover" must be a numeric field (e.g., data type is double) before using the Data Cleansing tool. Use a Select tool to change data type as necessary.*)

i. Examine the relation between forced turnover and the three variables you created earlier, i.e., return-on-assets, liability-to-assets ratio, and R&D-to-assets ratio. In particular, your analysis should include:
(1) Group records by forced turnover (i.e., turnover = 1 and = 0, respectively), then separately calculate the means (i.e., averages) for the three variables.
(2) correlations between forced turnover and the three variables.
(3) scatter plots showing the relation between forced turnover and the three variables (the plots might look a little strange as the dots will form two lines).

From here, create a new Word document to discuss the results. In particular, the Word file should start with screenshots of your workflow(s) and the results. **If you plan to include all analyses in a single workflow, you can wait until the end to take a screenshot (you don't need to provide screenshots for each step).** Please make sure the screenshots are legible. Below the screenshots, discuss your findings, focusing on whether bad performance (as proxied by low ROA, high liability-to-assets ratio, and low R&D) is related to CEO's forced turnover.

*****MSA students start from here*****

j. For i(1), further test if the averages are statistically different. Report and discuss your results in the Word document.

k. Estimate the following logistic regression:

$$\Pr(turnover = 1) = Logit(\beta_0 + \beta_1 ROA + \beta_2 lnAssets + \beta_3 LTA + \beta_4 RND)$$

Again, in the Word document, report and discuss your findings. Do the findings make intuitive sense? Are the findings consistent with those from step j?

l. Create a neural network to predict CEO turnover, using the same set of predictors as in k. Then compare it with the logistic model (no need to use the complex method such as a holdout sample). Which model performs better?

m. Suppose there is a firm with the following characteristics:
ROA = -0.2, lnAssets= 5, LTA=0.8, and RND=0.05. Predict the probability of a forced CEO turnover using the better-performing model in step l.

Is this probability higher or lower than the average turnover of the sample?

Based on the findings above, if you were a consultant to the CEO, do you think the CEO should be concerned about losing their job? What would your recommendation be to help the CEO better secure their job?

n. Submit the Word document and the Alteryx workflow file to Canvas (note the workflow file has a file extension of .yxmd. Please do NOT submit the Alteryx database file).